

Mohammad Hasnain Raza

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EDUCATION

- **University of Southern California** Los Angeles, CA
Master of Science in Computer Science; GPA: 4.00
◦ Coursework: Algorithms, Database Systems, Programming System Design, Computer Networks, Machine Learning
- **RV College of Engineering** Bengaluru, India
Bachelor of Engineering in Aerospace Engineering; GPA: 3.86, Silver Medalist
◦ Coursework: Engineering Mathematics, C Programming, Computational Fluid Dynamics (CFD), Scientific Computing

EXPERIENCE

- **Deloitte** Bengaluru, India
Technology Analyst
Aug 2022 – Nov 2024
◦ Improved system throughput by 10x by automating manual onboarding with Pega PRPC workflows, implementing rule-based routing across tax, credit, and legal approval chains for customer creation and modification requests across 35+ countries.
◦ Built a REST API orchestration layer connecting Pega with Informatica MDM and downstream SAP systems, achieving sub-2s latency and ensuring consistent propagation of validated master records across distributed enterprise systems.
◦ Built LLM-assisted categorization workflows using few-shot prompting to classify customer data requests, achieving 92% agreement with human annotations through iterative refinement using human feedback.
◦ Implemented data preprocessing pipelines for LLM-based workflows, including automated quality checks, Z-score based outlier filtering, and PII masking, reducing data preparation effort by 30%.
- **Defence Research and Development Organisation (DRDO)** Bengaluru, India
Research Intern
Jan 2022 – Aug 2022
◦ Led a 4-member team to automate transient CFD simulations using PyFluent (k- ω SST, overset mesh), generating 1.2TB+ of aerodynamic data across multiple release conditions for downstream modeling and analysis.
◦ Engineered feature sets from pressure and force coefficients and applied multi-variate regression to model store separation trajectories, identifying key predictors influencing stability under varying flow conditions.
◦ Built data preprocessing and post-processing pipelines using Python and MATLAB, including Z-score based outlier filtering and time-series feature extraction, enabling reliable inputs for predictive modeling and downstream validation against sensor data.
- **Prana.ai** Remote
Founding Engineer
Sep 2019 – Dec 2021
◦ Built end-to-end ML pipelines for 5M+ MRI/CT volumes, including normalization, patch extraction, and class-balanced sampling, enabling efficient training of 3D deep learning models on large-scale medical datasets.
◦ Designed and trained depthwise separable 3D CNNs for medical image segmentation, achieving < 0.8s inference latency through architecture optimization and efficient model deployment for real-time diagnostic use.
◦ Developed SO(3)-equivariant CNN-based super-resolution models using the e3nn library, leveraging rotation-equivariant convolutions to improve volumetric image quality by 35% over baseline methods.

KEY PROJECTS

- **Project Vimaan – AI Voice Command NLU System** Los Angeles, CA
NLP/ML Pipeline with X-Plane Integration
Sep 2025 – Present
◦ Built a data-generation pipeline producing 30,000+ labeled voice commands for joint intent and slot classification, using schema-driven templates and LLM-based paraphrasing (Pegasus, FLAN-T5) to improve linguistic diversity and label consistency.
◦ Fine-tuned a DistilBERT model for joint intent classification and entity extraction, optimizing hyperparameters and label design, and applied INT8 quantization to reduce model size by 4x while maintaining 98% accuracy.
◦ Deployed the model within the X-Plane simulator using a plugin-based architecture with UDP-based IPC, enabling real-time inference with sub-500ms latency and integrating TTS feedback for closed-loop command validation.
- **USC Ledger – AI-Powered Financial Management Platform** Los Angeles, CA
LLM Integration & Data Engineering
Aug 2025 – Present
◦ Built a transactional data pipeline (React, Node.js, MongoDB) with a sequential reconciliation engine to ensure 100% consistency across concurrent updates, producing reliable structured data for downstream analytics and modeling.
◦ Engineered feature processing logic using fixed-precision arithmetic and 800ms debounced writes to handle high-frequency financial events, reducing API load and ensuring stable inputs for real-time inference.
◦ Integrated LLM-based analytics using the Google Gemini API, designing prompt templates and context summarization to detect spending anomalies and generate structured financial insights from transaction data.

SKILLS

- **Languages:** Python (Advanced), SQL, R, MATLAB, Java, C/C++, JavaScript
- **Machine Learning:** Classification, Regression, NLP, Feature Engineering, Model Training, Evaluation, Hyperparameter Tuning
- **Frameworks:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, Pandas, NumPy
- **Data & Systems:** Data Pipelines, ETL, Experimentation, Statistical Analysis
- **LLM & NLP:** Transformer Models, BERT, T5, Prompt Engineering, LLM Fine-tuning
- **Tools:** Docker, AWS, Git, Tableau, Model Quantization, Model Monitoring